



TMNA IT Experiences Process

DATA ENABLEMENT AND GLOBAL COMPLIANCE DATA SECURITY PROCESS

Purpose The purpose of the Data Governance Security process is to document a set of guidelines for ensuring proper management of enterprise wide information including the Toyota Big Data Platform.

Data security processes and procedures will ensure that the right people can use and update data in the right way, and that all inappropriate access and updates are restricted. An effective data security management function establishes judicious governance mechanisms that are easy enough to abide by on a daily operational basis by all stakeholders.

Data Enablement and Global Compliance is the group that manages Data Governance at Toyota Motor North America (TMNA). The mission is to empower Toyota through effective usage of superior quality and highly secured data assets.

The goal is to deliver a Data Enablement and Global Compliance framework for TMNA to enable trusted, consistent and well-defined information which drives smarter business decisions across the entire information supply chains.

Data Security Management is the planning, development and execution of security processes and procedures to provide proper authentication, authorization, access and auditing of information assets. Data Stewards will guide all the data management functions that are performed under the Data Enablement and Global Compliance program.

Scope The scope of this Data Governance Security Process document is at the level defined below:

- Define the roles and responsibilities for different data usage and establish clear lines of accountability
- Develop best practices for effective data management and protection
- Protect the data against internal and external threats (e.g. breach of privacy and confidentiality)
- Ensure that data audit trails are effectively documented within the processes associated with accessing, retrieving, reporting, managing and storing of data
- Establish a culture of Data Governance across the IS organization where there is zero tolerance for unauthorized access.

- Avoid negative exposure, fines and lost revenue associated with data breaches
- Maintain compliance with complex regulations during audits
- Protect proprietary information against security threats caused by enhanced mobility and new communication channels
- Protect against theft and accidental disclosure of sensitive information by employees and partners
- Reduce the risk and business impact by restricting access at the data level.
- Classify data, build processes, adopt technologies and implement controls based on how organization uses data

Compliance Compliance with this process is mandatory. When not meeting this process' requirements for business reasons, a TMNA manager must seek and obtain an approved Exception, requested and documented through the Exception process in [Toyota's Ticketing Systems](#). Team Members who violate this process may be subject to corrective action, up to and including termination. You are subject to all agreements you signed with the Company, as well as all Company rules and policies, including but not limited to the [TMNA Policies](#).

If you see or hear of an activity that may violate this process, whether by a Team Member, Corporate Partner, or Third-Party; you must report that activity. If this is a security incident you can email Security@toyota.com or call the Help Desk at (888) 488-4357. Reports may be made anonymously via the Toyota Speak Up Line by calling (844)SPEAK-99 or by visiting the [Toyota EthicsPoint website](#).

Owner	DataEnablement@toyota.com
Approver	DataEnablement@toyota.com
Initial Approval Date	08/30/2017
Effective Date	08/30/2017
Revision Date	10/15/2021
Last Reviewed Date	10/15/2021
Revision Number	1.0
SharePoint Location	DE Microsite (toyota.com)

NEVER USE PRINTED COPIES OF ANY PROCESS; USE SHAREPOINT VERSION

This document contains the following sections:

1.	Processes.....	3
2.	Applications hosted on Cloud	4
3.	Data Process Exceptions	6
4.	Roles and Responsibilities.....	6

5.	References, Regulations, and Guiding Standards	7
6.	Data Security Lifecycle	8
7.	Data Security Dimensions	8
8.	Data Security Framework	9
9.	Active Directory Groups	9
10.	Data Security, Process Change	10
11.	Revision History	10
12.	Appendix for Relevant Processes	11

1. Processes

Item	Requirement	Ref/Flag
1	Data Security framework will be applied for all the data on-boarded to TBDP.	
2	All the data on-boarded on TBDP should be classified for confidentially at a table, view and attribute level. Data classified as confidential, highly confidential, or legal in TBDP should be encrypted, tokenized or masked as per the requirement.	
3	Access to data on TBDP will be granted only through Active Directory (AD) groups. Data security team will create AD groups and will assign privileges to TBDP users by aligning them to appropriate role groups. No user should be granted access to TBDP outside an AD group.	
4	All the Active Directory (AD) groups created for providing TBDP access to users should be reviewed and approved by Data Stewards.	
5	All the data access requests for TBDP should have a One Toyota service (1TS) request number associated with it. No access should be granted to any user by e-mail, messenger or phone call request.	

Item	Requirement	Ref/Flag
6	Data Security audits (internal/external) should be carried out on predefined interval to ensure the organization is complaint with regulatory requirements. TBDP data should be reviewed for its reliability and accuracy to analyze the effectiveness and maturity of data security processes and practices.	

2. Applications hosted on Cloud

The following set of processes apply to applications/data hosted on cloud

Item	Requirement	Ref/Flag
1	Need to ensure confidential and critical data is restricted to people those are not entitled to use it. Access to this data is tightly controlled with only a few individual users being entitled to see or use the data. Critical data is generally stored in purpose built applications, often in an encrypted format, even within internal systems.	
2	Need to ensure that all the information is the subject of a data protection or Freedom of information access request. All information held in cloud held by the organization and therefore may be the subject of a data protection or Freedom of information access request	
3	Need to ensure data ownership roles in data classification efforts and their respective rights. It's important to establish a clear custodial chain of ownership for all data assets. The data classification process document identifies different data ownership roles in data classification efforts and their respective rights.	
4	Need to ensure reclassifying or changing the classification state of a data asset needs to be done when a user or system determines that the data asset's importance or risk profile has changed. Reclassifying or changing the classification state of a data asset needs to be done when a user or system determines that the data asset's importance or risk profile has changed. This effort is important for ensuring that the classification status continues to be current and valid. Most content that is not classified manually can be classified automatically or based on usage by a data custodian or data owner.	

Item	Requirement	Ref/Flag
5	Authentication: Following authentication techniques/methods needs to be used for data hosted on cloud. i) Need to enable MFA (Multi Factor Authentication) ii) Need to enable and use IAM (Identity Access Management) accounts for day to day activities iii) Need to organize IAM users into IAM groups iv) All the IAM groups created should have dedicated security policies. v) Need to use multiple accounts for different geographical regions and department/business units	
6	Following network security processes apply for Data in Motion i) Need to use TLS (Transport Layer Security)/SSL (Secure Sockets Layer) to provide communication security over network. ii) Need to enable VPN's (Virtual Private Networks), Direct Connect and network ACL's (Access Control List) iii) Need to use WAF (Web Application Firewalls) for HTTP conversations to stop common attacks like ▪ Cross-site scripting(XSS) ▪ Cross-site request forgeries (CSRF) ▪ Denial of Service (DoS) ▪ SQL Injection iv) Need to enable WAF (Web Application Firewall) that looks details into packet payloads, protocols unlike other firewalls which looks only at header details	
7	Authorization: Following authorization techniques/methods needs to be used for data hosted on cloud. i. Need to integrate Cloud Active Directory with Enterprise Active Directory ii. Need to avoid using local accounts and will require necessary approvals in case of exceptions iii. Always keep data storage objects as private	
8	Data Protection: Following Data Protection techniques/methods needs to be used for data hosted on cloud. i. Need to enable cross-region replication so that each object gets replicated across all availability zones ii. Need to use industry standard RSA public/private key pairs. iii. Do not store private key on Cloud instances. iv. Need to use 256 byte encryption key for encrypting the DB Instances, snapshots, backups and logs v. KMS keys are tied to a specific region which means encrypted instances or snapshots can be replicated across regions	

3. Data Process Exceptions

Item	Requirement	Ref/Flag
1	Data outside of the TBDP data security framework may not have this process applied to them	
2	Data that is Non-Sensitive on-boarded on TBDP can be grouped per domain under a combined Active Directory group (AD) and may or may not be encrypted, tokenized or masked.	

4. Roles and Responsibilities

Implementation and administration of data security process is primarily the responsibility of security administrators.

Role	Responsibility
Data Stewards	• Data Stewards are responsible for promoting and fostering a data security culture for on-boarding applications in TBDP environment • Ensuring data security measures are carried out for every data collection under their area of responsibility. • Data Stewards are responsible for evaluating and determining the appropriate confidentiality level for data.
Data Security Lead	▪ Interacting with stakeholders' teams and understanding data and its classification. ▪ Identify, create and update AD (Active directory) groups for granting user data access ▪ Detailed requirements gathering and design definition for the data security framework ▪ Provide technical guidance for implementing data security measures and guidelines ▪ Ensure data security thresholds are defined and met, drive plans and remediation activities where thresholds are not met

Role	Responsibility
Requestor	▪ Define the purpose of the request and provide business justification ▪ Provide all the mandatory information and clarification
Business/Technical Owner	▪ Analyze and provide Data Security requirements for onboarding application ▪ Assist Data Stewards in classification of data
Data Security Developer	▪ Design and develop security rules in coordination with Data Security lead. ▪ Perform necessary unit testing and support in deploying scripts, UAT and PROD environment (as needed). ▪ Resolve the support issues or perform the support tasks on different environments ▪ Monitor issues and highlight findings

5. References, Regulations, and Guiding Standards

- Please follow the link for Information Security - GOVERNANCE, RISK AND COMPLIANCE Policy
[TMNA Information Security - GRC Policy](#)
- Please follow below link for security policy on Cryptography.
[TMNA Information Security Policy - CRYPTOGRAPHY](#)
- Please refer below link for more security related policies
<https://myteams.toyota.com/sites/tmnaisinfosecGRC/Policies/SitePages/Home.aspx>
- Please refer the below link to find more details on TMNA Data Encryption Policy.
[TMNA Data Encryption Policy](#)

6. Data Security Lifecycle

Data Security lifecycle	Description
Identify	- Identify data assets and its sources - Understand business, risks and liabilities - Understand the classification of data (Public, Confidential, etc)
Detect	- Define guidelines, authorization policies and controls - Create AD (Active Directory) groups for access control.
Protect	- Block unauthorized access to sensitive data - Prevent data from security breach

7. Data Security Dimensions

Data Security Dimension	Description
Privacy	The Data Enablement and Global Compliance team will secure the data on data platforms, and will maintain its confidentiality. The data collected, processed and integrated will be shared only with authorized personnel
Protection	Sensitive data will be safeguarded against internal and external threats. Data will be classified, encrypted and decrypted to avoid data theft. All the user activities will be tracked, audited and controlled.

Antitrust

With the launch of a various joint ventures between Toyota Motor North America and other automotive manufacturers, additional data security considerations regarding information that falls under the Federal Antitrust regulatory scope must be taken. TMNA applications that have data considered antitrust (see below), must comply with antitrust law which extends to anywhere that dataset is replicated.

As these datasets are advertised in the Data Marketplace, they will be clearly metadata tagged ("Antitrust" label on that product) to indicate they contain elements considered under the Antitrust constraints.

These include but are not limited to the following types of information:

- Marketing and sales plans of any vehicle manufactured as a result of the joint venture
- Projected or proposed wholesale or retail prices for any vehicle or vehicle part sold to third parties manufactured by the joint venture
- Supplier negotiations, information, contracts and/or cost data unnecessary to the formation or operation of the joint venture
- Wages, benefits, or similar information about employee compensation as well as non-public aspects of hiring practices or processes

- Any information not made public including prices paid for components/parts, other cost information, production plans/volumes, future product planning, product lineups or specifications, sourcing plans regarding vehicle models not being manufactured by the Joint Venture

Additional points of clarification:

- Future planned/forecasted manufacturing production volumes for the joint venture manufacturing facilities
- Parts pricing for parts added to the vehicle during the manufacturing process (aka direct parts)
- New model parts lists and advanced technology capabilities that have not been publicly released

For a TMNA resource to be considered for access to datasets that include Antitrust data, requesters of this access must complete the joint venture specific antitrust training **prior** to submitting a Data Access Request via Data Marketplace, and provide a valid business justification for needing access to the data.

New hires at the joint venture plants will receive antitrust training automatically. If you are not for sure which application datasets require training or you are unsure which training to complete, please contact the data steward responsible for the application's domain. Provided below is the link to help determine which data steward to contact.

[Application List by Domain.pdf \(toyota.com\)](#)

8. Data Security Framework

Data Security Framework on big Data platform

- Authentication:** (Validate users are who they say they are) Toyota big data platform (TBDP) should be integrated with Active Directory to restrict unauthorized users.
- Authorization:** (Identify the right individuals and grant them the right privileges to specific, appropriate views of data) User Authorization should be enabled on TBDP to allow developers and business stakeholders to gain access to the data they **need**, but not a byte more.
- Data Protection:** (The process of safeguarding important information from corruption or loss). File Level Encryption should be enabled on TBDP for highly Confidential data or above. If data is stored in a Cloud hosted infrastructure, encryption may apply to more levels (e.g. Protected). The encryption process applies to data in motion, and at rest.
- Data Audit:** (Review security actions and user activity to ensure compliance with regulations and conformance with process and standards) User and Data Auditing capabilities should be enabled on TBDP.

9. Active Directory Groups

The below diagram provides details on the domains which will guide security team in creating Active Directory (AD) groups.

e.g.:

- AD group example - TBD<Environment>_<Domain>_<Subdomain>_<data classification>
- TBD - Toyota Big data Platform
- Environment – DEV, QA, PROD
- Domain, Subdomain - can be seen in the diagram below
- Data Classification – Please refer data classification process document
- Constraint – AD group have a constraint on naming convention that it should be only 23 bytes



10. Data Security, ProcessChange

Changes to the process shall be reviewed and approved by Data Stewards and Data Security Owner. Communication of Process Changes will be defined by TMNA processes for Team Members and TMNA's Suppliers.

11. Revision History

Version	Date	Author	Description	Section(s) Affected
1.0	08/30/2017	DataEnablement@toyota.com	The Data Security Process (the "Process") ensures that the right users use & update data	All

Version	Date	Author	Description	Section(s) Affected
			the right way, and that all inappropriate access and updates are restricted	
1.0	10/15/2021	DataEnablement@toyota.com	Added process in the ITx template	All

12. Appendix for Relevant Processes

[TMNA Data Enablement Data Quality Process](#)

[TMNA Data Enablement Metadata Process](#)